

b. Analyze customer data to identify trends and patterns that can be used for business decision-making.

```
#Import Required Libraries
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
#Load dataset
```

```
df = pd.read_csv("Retail_sales.csv")
```

Analysis 1 - Product category analysis

```
print(df["Product Category"].value_counts())
```

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

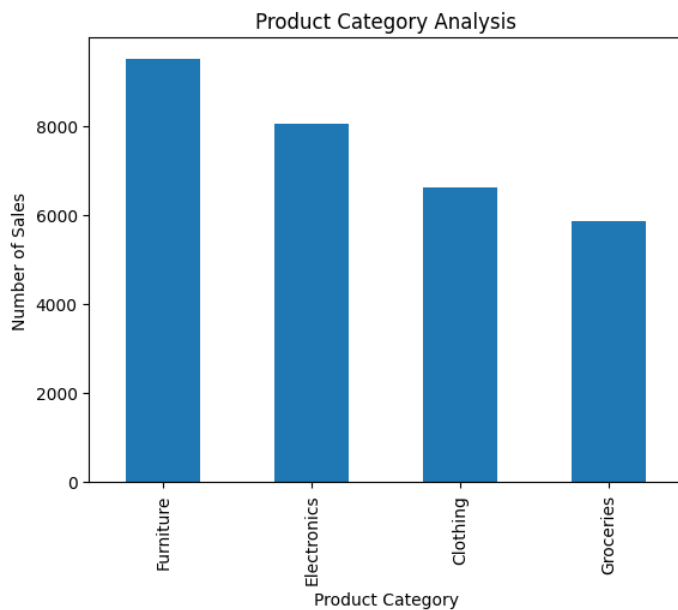
```
df["Product Category"].value_counts().plot(kind="bar")
```

```
plt.title("Product Category Analysis")
```

```
plt.xlabel("Product Category")
```

```
plt.ylabel("Number of Sales")
```

```
plt.show()
```

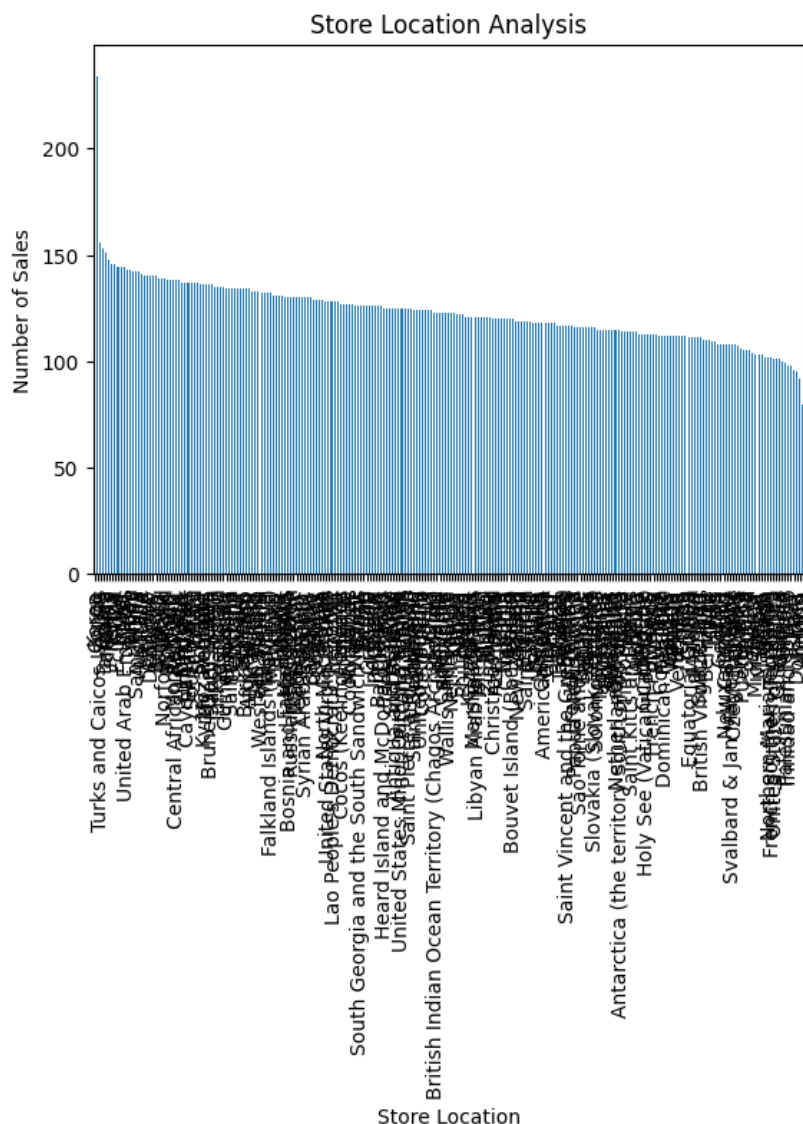


Analysis 2 - Store Location Analysis

```
print(df["Store Location"].value_counts())
```

```
Store Location
Korea      237
Congo      234
Turks and Caicos Islands  156
Guam       153
Tanzania   151
...
Trinidad and Tobago  98
Haiti       96
Dominica    95
Ireland     92
Andorra     80
Name: count, Length: 243, dtype: int64
```

```
df["Store Location"].value_counts().plot(kind="bar")
plt.title("Store Location Analysis")
plt.xlabel("Store Location")
plt.ylabel("Number of Sales")
plt.show()
```

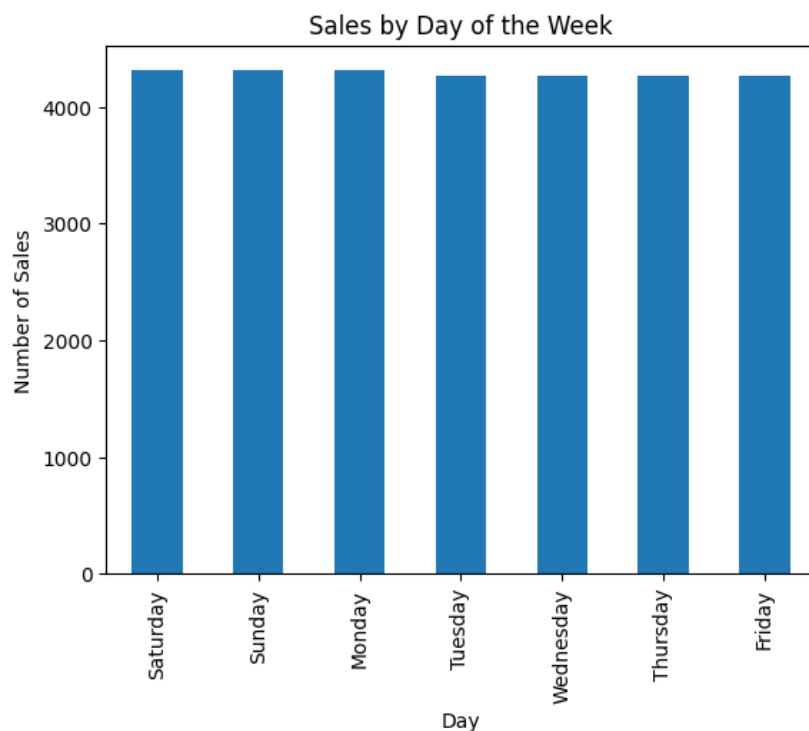


Analysis 3 - Day of the Week Analysis

```
print(df["Day of the Week"].value_counts())
```

```
Day of the Week
Saturday      4310
Sunday        4309
Monday        4309
Tuesday       4268
Wednesday     4268
Thursday      4268
Friday        4268
Name: count, dtype: int64
```

```
df["Day of the Week"].value_counts().plot(kind="bar")
plt.title("Sales by Day of the Week")
plt.xlabel("Day")
plt.ylabel("Number of Sales")
plt.show()
```

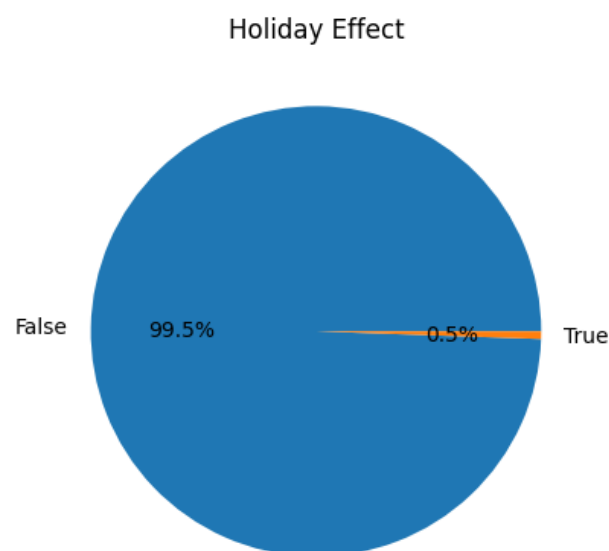


Analysis 4 - Holiday Effect Analysis

```
print(df["Holiday Effect"].value_counts())
```

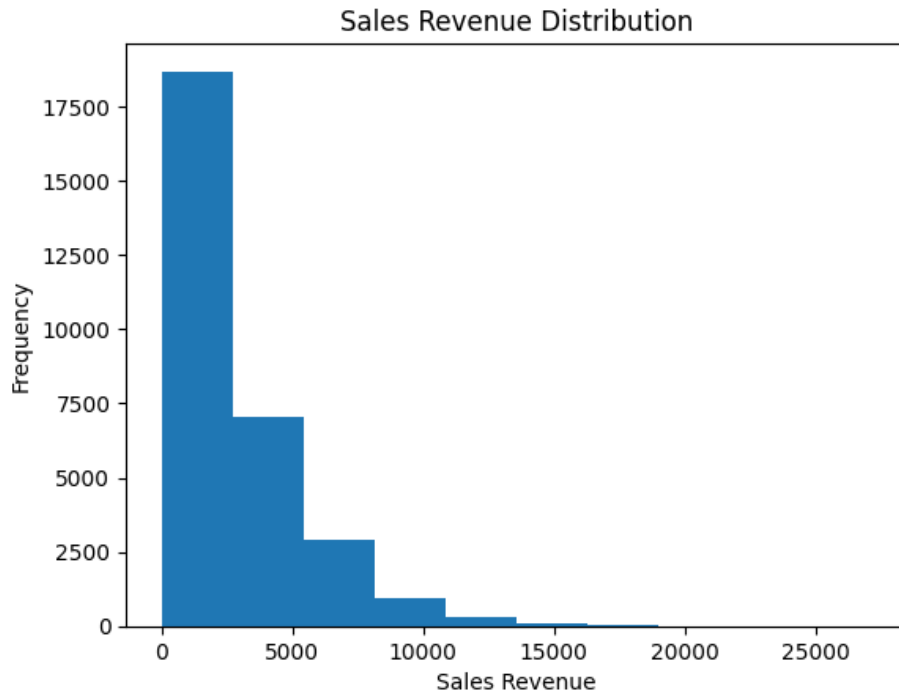
```
Holiday Effect
False    29836
True      164
Name: count, dtype: int64
```

```
df["Holiday Effect"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Holiday Effect")
plt.ylabel("")
plt.show()
```



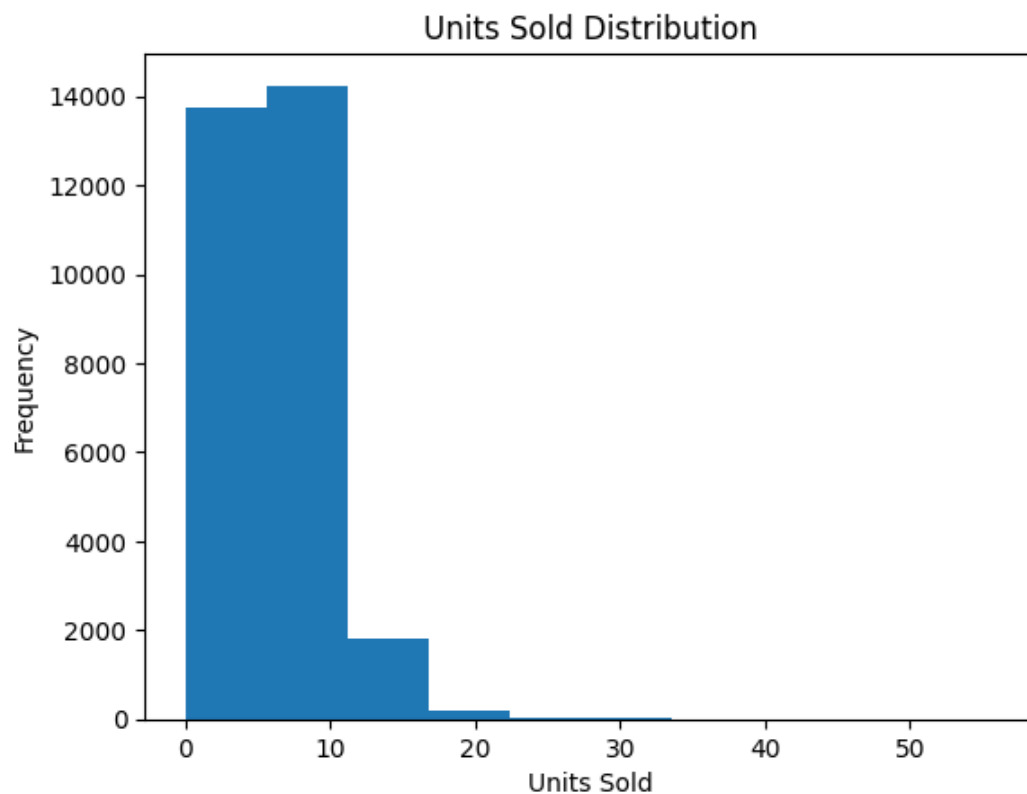
Analysis 5 - Sales Revenue Analysis

```
plt.hist(df["Sales Revenue (USD)"], bins=10)
plt.title("Sales Revenue Distribution")
plt.xlabel("Sales Revenue")
plt.ylabel("Frequency")
plt.show()
```



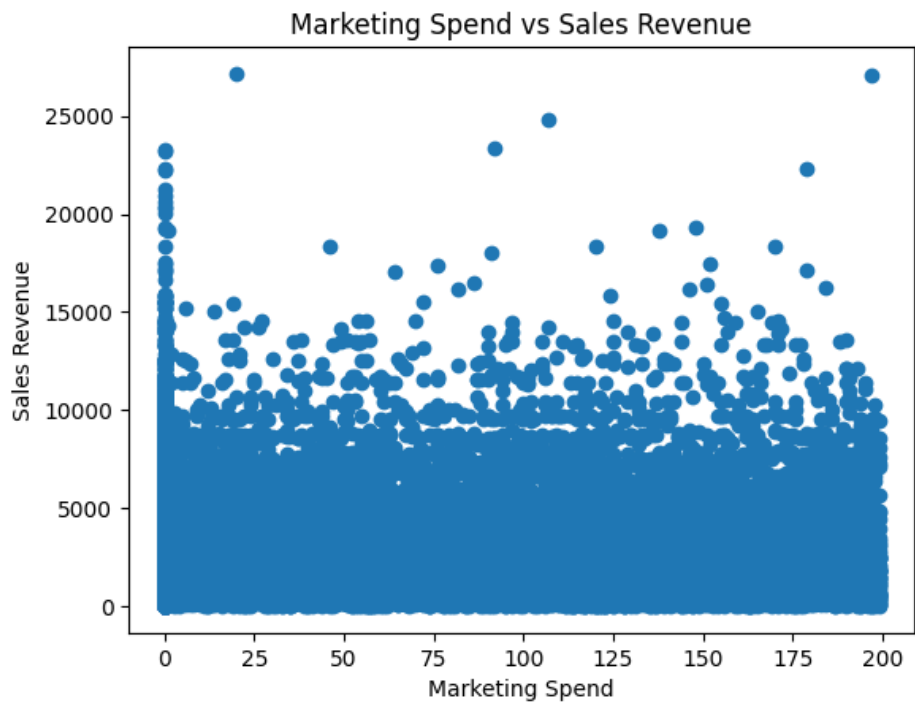
Analysis 6 - Units Sold Analysis

```
plt.hist(df["Units Sold"], bins=10)
plt.title("Units Sold Distribution")
plt.xlabel("Units Sold")
plt.ylabel("Frequency")
plt.show()
```



Analysis 7 - Marketing Spend vs Sales Revenue

```
plt.scatter(df["Marketing Spend (USD)", df["Sales Revenue (USD)"])
plt.title("Marketing Spend vs Sales Revenue")
plt.xlabel("Marketing Spend")
plt.ylabel("Sales Revenue")
plt.show()
```



Analysis 8 - Correlation Analysis

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Oranges")  
plt.title("Correlation Heatmap")  
plt.show()
```

